Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/20/2024
PI Name:	Bailey, Susan M. Ph.D.		
Project Title:	Telomeres and the One Year Mission Project		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
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COI Name (Institution):	Jeevarajan, Antony Ph.D. (NASA Johnson Space Center)		
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Task Description:	The ultimate goal of the studies proposed here is to establish temporal profiles of human telomere length dynamics and DNA damage responses of importance for maintenance of human health and performance during long-duration deep space missions. We hypothesize that telomere length dynamics (changes over time) represent a particularly relevant and informative biomarker of health for the astronauts, as it reflects the combined experiences and exposures encountered during spaceflight. That is, an astronaut's individual genetic susceptibilities, unique lifestyle stresses encountered (e.g., nutritional, psychological, physical), and particular environmental exposures (e.g., altered atmospheres, microgravity, space radiations) are all integrated and captured as changes in telomere length. Thus, the rate at which telomeres shorten provides a general measure of health that can be linked to aging, as well as to risk of developing age-related pathologies, ranging from reduced immune function and dementia, to cardiovascular disease and cancer. Importantly, functional telomeres are also essential for maintaining genomic integrity and stability, as they protect chromosomal termini from inappropriate degradation, and prevent these natural DNA ends from being recognized as broken DNA and triggering inappropriate DNA damage responses (DDRs). To identify trends in adaptations to human health and performance during long-duration low-Earth orbit, we propose telomere length and DDR/cytogenetic measures pioneered and validated in the NASA Twins Study/first One Year Mission, across the Integrated One-Year Mission Project onboard the International Space Station and the concurrent ground analog (prolonged isolation) component.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Identifying interactive effects of genetic and nongenetic telomere length determinants and DDRs will improve understanding of aging and aging trajectories (disease risk), as well as guide future studies and development of mitigation strategies/countermeasures for improving health-span, not only in astronauts on long-duration missions, but for those on Earth, as well.
Task Progress:	Telomeres, nucleoprotein complexes that “cap” the ends of linear chromosomes, are composed of highly conserved, tandem arrays of G-rich repetitive sequence. Telomere length is an inherited trait that varies considerably between individuals, ranging from ~ 5 to 15kb in humans. Functional telomeres are essential for maintaining genome integrity and stability, as they protect chromosomal termini from inappropriate degradation and prevent these natural DNA ends from being recognized as broken DNA (double-strand breaks; DSBs) and triggering of inappropriate DNA damage responses. It has long been appreciated that telomere length erodes with cellular division due to the end-replication problem, causing them to shorten until reaching a critically short length, at which point a permanent cell cycle arrest known as cellular senescence is entered and cells stop dividing. Preservation of telomere length requires Telomerase, the specialized RNA-dependent (TERC) reverse transcriptase (TERT) capable of maintaining telomere length via de novo template-mediated addition of telomeric repeats onto the ends of newly replicated chromosomes. However, telomerase activity is repressed in most normal human tissues (thus, telomeres shorten and cells senescence; an effective tumor suppressor), and is prominent only in highly proliferative populations like germ-line, stem, and the vast majority of cancer cells (acts as an oncogene), thereby endowing them with extended or unlimited replicative potential. While it is well established that telomere length diminishes with normal aging and oxidative stress, it is becoming increasingly appreciated that telomere length is influenced by a variety of other factors as well, including biological sex, diet, lifestyle factors [e.g., smoking and obesity, physical activity, psychological stress], chronic stress and disease. Telomerase activity is also influenced by comprehensive lifestyle changes, acute psychological stress, and environmental exposures. Taken together, telomere length maintenance becomes an informative measure of general health, in that it represents a key integrating component for the cumulative effects of genetic (individual susceptibilities), environmental (air pollution, ionizing radiations), and lifestyle (nutritional, psychological, physical) factors on aging and age-related diseases. The rate at which telomeres shorten or change over time (telomere length dynamics) provides an informative biomarker of aging that can also be linked to risk of age-related degenerative pathologies, ranging from reduced immune function [including increased risk of the common cold], loss of fertility, idiopathic pulmonary fibrosis, and dementias, to cardiovascular disease (CVD) and cancer. Importantly, recent evidence supports telomere length not only as an informative biomarker, but even a determinant of CVD and cancer, thereby suggesting a telomere length trade-off of sorts; on the one hand, shorter telomeres, senescence, and cancer resistance in exchange for increased risk of age-related (degenerative) CVD, and on the other, longer telomeres and increased cancer risk (proliferative pathology) for reduced CVD risk. Such health effects relevant to spaceflight are largely unknown and controversial, yet they have very real potential for influencing performance during long-duration missions. Longitudinal changes in telomere length maintenance in human populations are also not well understood, particularly as associated with long-duration spaceflight, with its unique lifestyle factors and environmental exposures. We speculated that telomere length dynamics (changes over time) represent an especially relevant and informative biomarker of health risk for astronauts, as it reflects the combined exposures and experiences encountered during spaceflight. That is, an astronaut’s individual genetic susceptibilities, unique lifestyle stressors (e.g., nutritional, physical, and psychological) and environmental exposures [e.g., microgravity and the space radiation environment, which includes galactic cosmic rays (GCR), solar particle events (SPEs), as well as secondary particles that arise from interactions with spacecraft shielding], are all integrated and captured as changes in telomere length over time. We had the remarkable opportunity to assess telomere length and telomerase activity in twin (~1year mission duration) and unrelated (~6-month mission duration) astronauts, and age/sex-matched ground control subjects: pre-flight (to establish baseline); during flight (to evaluate short-term/temporary changes); and post-flight (to evaluate long-term/permanent changes). Our ongoing work is demonstrating changes in human telomere length dynamics specifically associated with spaceflight, supporting telomeres as integrative biomarkers that encompass the extraordinary life stressors and environmental exposures encountered. Thus, telomeres are being established as informative biomarkers of astronaut health, disease risk, and aging. Individual differences are also being observed, therefore determining responses in a larger more diverse population of astronauts as proposed as part of CIPHER represents a critical next step. Importantly, our previous results have uniquely poised and informed our current approaches aimed at improving understanding of the mechanisms underlying such dramatic changes in telomere length associated with spaceflight and exposure to the space radiation environment; specifically, preliminary evidence suggests and we are currently monitoring changes in cell population dynamics (e.g., stem/progenitor cells), and/or transient activation of the telomerase independent, recombination-mediated Alternative Lengthening of Telomeres/ALT pathway of telomere maintenance. Together with evidence of radiation-induced chromosomal/genome instability, such vital mechanistic insight will provide potential targets for the development of countermeasures, thereby benefiting and helping to enable success of future missions, as

	well as informing development of novel anti-aging strategies on Earth. Moreover, as Telomeres 2 and other CIPHER projects mature and complete (e.g., Vascular Aging) and more data becomes available, improved precision space medicine strategies will better inform health and aging trajectories for individual astronauts, and the view of space as a model for accelerated aging will be rigorously tested.
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